

```
%cd /content
!rm -rf PMUY-AI
!git clone https://github.com/tanisha0710-ui/PMUY-AI.git
%cd PMUY-AI
```

```
/content
Cloning into 'PMUY-AI'...
remote: Enumerating objects: 224, done.
remote: Counting objects: 100% (224/224), done.
remote: Compressing objects: 100% (207/207), done.
remote: Total 224 (delta 109), reused 37 (delta 7), pack-reused 0 (from 0)
Receiving objects: 100% (224/224), 223.74 KiB | 3.39 MiB/s, done.
Resolving deltas: 100% (109/109), done.
/content/PMUY-AI
```

```
!pip install pandas statsmodels pyarrow gdown
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: statsmodels in /usr/local/lib/python3.12/dist-packages (0.14.6)
Requirement already satisfied: pyarrow in /usr/local/lib/python3.12/dist-packages (18.1.0)
Requirement already satisfied: gdown in /usr/local/lib/python3.12/dist-packages (5.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.1)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.1)
Requirement already satisfied: scipy!=1.9.2,>=1.8 in /usr/local/lib/python3.12/dist-packages (from statsmodels) (1.13.1)
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from statsmodels) (0.5.6)
Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.12/dist-packages (from statsmodels) (25.0)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-packages (from gdown) (4.13.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from gdown) (3.29.0)
Requirement already satisfied: requests[socks] in /usr/local/lib/python3.12/dist-packages (from gdown) (2.32.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from gdown) (4.67.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil) (1.17.0)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4) (2.7)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4) (4.12.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests) (2025.1.31)
Requirement already satisfied: PySocks!=1.5.7,>=1.5.6 in /usr/local/lib/python3.12/dist-packages (from requests[socks]) (1.9.0)
```

```
!mkdir -p data
!gdown --id 1V94LK_vh0R-D3Hioa5J8hqzenECcuyCZ -O data/panel.parquet
```

```
/usr/local/lib/python3.12/dist-packages/gdown/__main__.py:139: FutureWarning: Option `--id` was deprecated
  warnings.warn(
Downloading...
From (original): https://drive.google.com/uc?id=1V94LK\_vh0R-D3Hioa5J8hqzenECcuyCZ
From (redirected): https://drive.google.com/uc?id=1V94LK\_vh0R-D3Hioa5J8hqzenECcuyCZ&confirm=t&uuiid=e416fc7
To: /content/PMUY-AI/data/panel.parquet
100% 203M/203M [00:02<00:00, 98.3MB/s]
```

```
!ls -lh data
```

```
total 194M
-rw-r--r-- 1 root root 194M May  5 10:53 panel.parquet
-rw-r--r-- 1 root root 339 May  6 13:10 README.md
```

```
!python main.py
```

Template outputs written to outputs/
Replace the placeholder fields before you submit the milestone or the final project.

```
!cat outputs/baseline_metric.json
```

```
{
  "metric_name": "naive_did_pp",
  "description": "Unadjusted DiD \u2014 2x2 weighted means, no controls, no FE. Treatment = High exposure",
  "treatment_group": "High exposure states (high_exposure=1, below NFHS-4 median) \u2014 PMUY-targeted",
```

```

"control_group": "Low exposure states (high_exposure=0, above NFHS-4 median) \u2014 control",
"treatment_pre_pp": 27.4,
"treatment_post_pp": 42.0,
"treatment_delta_pp": 14.6,
"control_pre_pp": 63.0,
"control_post_pp": 79.5,
"control_delta_pp": 16.6,
"value": -2.0,
"unit": "percentage points",
"note": "Naive DiD is negative (-2.0 pp): low-access states improved less than high-access states uncond",
"is_template": true
}

```

```
!cat outputs/milestone_manifest.json
```

```

{
  "milestone_date": "2026-05-06",
  "project": "The Impact of PMUY on Clean Fuel Adoption: A Difference-in-Differences Analysis Using NFHS-4",
  "team": [
    "Tanisha Aggarwal",
    "Neha Rana",
    "Jaswathi Lalitha R"
  ],
  "charter_locked": false,
  "status": "milestone",
  "sources": [
    {
      "name": "pmuy_data.csv",
      "file": "https://drive.google.com/file/d/1V94LK\_vh0R-D3Hioa5J8hqzenECcuyCZ/view",
      "rows_raw": 1238208,
      "rows_after_exclusions": 1235952,
      "nfhs4_obs": 600328,
      "nfhs5_obs": 635624,
      "states_uts": 35,
      "clean_fuel_mean_unweighted": 0.4565,
      "status": "verified",
      "probe_artifact": "artifacts/probes/replace_me_probe.md",
      "note": "Merged NFHS-4 and NFHS-5 household microdata. Primary outcome source."
    },
    {
      "name": "PPAC State-wise PMUY Connections",
      "url": "https://ppac.gov.in/consumption/state-wise-pmuy-data",
      "backup": "https://www.data.gov.in/resource/stateut-wise-number-pradhan-mantri-ujjwala-yojana-pmuy-connections",
      "status": "verified",
      "note": "Used only to validate high/low exposure split. Not a primary outcome source."
    }
  ],
  "treatment_definition": {
    "variable": "high_exposure",
    "direction": "1 = Low-access states (below NFHS-4 median) = PMUY-targeted",
    "nfhs4_median_pp": 52.3,
    "n_treatment_states": 17,
    "n_control_states": 18,
    "treatment_states": [
      "bihar",
      "jharkhand",
      "odisha",
      "meghalaya",
      "chhattisgarh",
      "assam",
      "west bengal",
      "madhya pradesh",
      "tripura",
      "rajasthan",
      "lakshadweep",
      "uttar pradesh",
      "nagaland",
      "himachal pradesh",
      "manipur",
      "arunachal pradesh",
      "uttarakhand"
    ]
  }
}

```

```
!cat outputs/primary_metric.json
```

```
{
  "metric_name": "did_coefficient_pp",
  "value": -2.0,
  "ci_lower": null,
  "ci_upper": null,
  "unit": "percentage points",
  "threshold": 2.0,
  "threshold_rule": "|value| >= 2.0 pp AND CI excludes zero",
  "passed": true,
  "notes": "Preliminary value set to naive DiD (-2.0 pp) as placeholder. Will be replaced with TWFE coeffi",
  "is_template": true
}
```

```
%cd /content/PMUY-AI
```

```
!grep -R "replace_me\\|blocked\\|is_template.*true" outputs || true
```

```
/content/PMUY-AI
outputs/baseline_metric.json: "is_template": true
outputs/primary_metric.json: "is_template": true
outputs/milestone_manifest.json: "probe_artifact": "artifacts/probes/replace_me_probe.md",
outputs/milestone_manifest.json: "Data probe: artifacts/probes/replace_me_probe.py"
```

```
!git status
!git add outputs/baseline_metric.json outputs/milestone_manifest.json outputs/primary_metric.json
!git status
```

```
On branch main
Your branch is up to date with 'origin/main'.
```

```
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   outputs/baseline_metric.json
    modified:   outputs/primary_metric.json
```

```
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  data/panel.parquet
```

```
no changes added to commit (use "git add" and/or "git commit -a")
On branch main
Your branch is up to date with 'origin/main'.
```

```
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   outputs/baseline_metric.json
    modified:   outputs/primary_metric.json
```

```
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  data/panel.parquet
```

```
!git config --global user.name "Tanisha Aggarwal"
!git config --global user.email "YOUR_EMAIL_HERE"
```

```
!git commit -m "Update milestone outputs with PMUY NFHS values"
```

```
[main 9d32f7f] Update milestone outputs with PMUY NFHS values
2 files changed, 3 insertions(+), 2 deletions(-)
```

```
TOKEN = "ghp_6EnQQfWiJCfto1ioBlnqrgwVpy0wNG18xP0B"
```

```
!git remote set-url origin https://{TOKEN}@github.com/tanisha0710-ui/PMUY-AI.git
!git push origin main
```

```
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 479 bytes | 479.00 KiB/s, done.
```

```
Total 5 (delta 4), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
To https://github.com/tanisha0710-ui/PMUY-AI.git
d70f53c..9d32f7f  main -> main
```